

DELCODE-PET | PET DICOM images

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Purpose:

This document provides background information regarding PET imaging data that can be requested from the DELCODE study.

Note: Publications using DELCODE-PET Florbetaben PET data should mention in the Acknowledgments/ Sponsoring section: *“Preparation of the Florbetaben data in this study was in part performed under a research agreement with Life Molecular Imaging. Some Florbetaben PET examinations were performed with support of the AMYPAD research consortium.”*

Available PET imaging data

DELCODE-PET includes PET examinations of **cerebral glucose metabolism** using ^{18}F -Fluorodesoxyglucose (FDG), and **brain amyloid burden** using NeuraceqTM (18F-Florbetaben [FBB]: Life Radiopharma, Berlin).

Generally, PET examinations were **only collected from Subjective Cognitive Decline (SCD) participants from the DELCODE study** who agreed to these additional examinations.

No complementary PET data are available from healthy controls, MCI and AD dementia participants!

For both FBB and FDG, PET examinations were **performed up to two times**: A baseline scan (“P0”) and a follow-up scan (“P24”).

Note: Due to the delayed start of DELCODE-PET, the baseline scan was not always performed at the baseline visit of the DELCODE main study, but during follow-up visits of the DELCODE main study. Accordingly, the “P0” and “P24” labels for the PET data are not equivalent with the “M0, M12, M24, ...” labels found in the MRI data. Moreover, while follow-up PET scans were originally scheduled for a 24-month delay (hence: “P24”), this was later extended to a time window of up to 36 months.

Data acquisition:

PET data acquisition followed established standard protocols. While there were some tracer-specific differences in preparation and tracer application, PET image acquisition was technically similar:

	FDG	FBB
Requested time since last food intake	≥ 6 hours	Not applicable
Maximum blood glucose level	≤ 140 ng/ml	Not applicable
Applied tracer activity	170-180 Megabequerel (MBq)	260-300 Megabequerel (MBq)
Rest period after injection	20 min	Not applicable
Start of emission scan	40 min post-injection	90 min post-injection
Duration of emission	20 min	
Acquisition mode	3D Listmode, <i>post hoc</i> reconstruction in 4*5 min time frames	
Reconstruction algorithm	<i>Iterative:</i> Based on established brain protocols at PET study sites • <i>Siemens + GE:</i> Ordered Subset Expectation Maximization (OSEM) • <i>Philips:</i> Line-of-response row-action maximum likelihood (LOR-RAMLA)	
Image corrections	• Decay • Randoms • Dead time • Scatter • Attenuation • Normalization	
Attenuation correction	Based on vendor-specific attenuation coefficient (μ) maps • <i>Siemens/ GE/ Philips PET/CT systems:</i> Derived from low-dose CT • <i>Siemens PET/MR systems:</i> Derived from DIXON-VIBE MR sequence	

Quality assessments:

After data transfer, PET images were inspected by the PET Group at the DZNE Bonn to check compliance with the acquisition protocol. This included:

- Check for completeness of the relevant image series
- Extraction of information from the reconstructed PET DICOM series
- Check of consistency of the available information with standard acquisition protocol
- Visual image check for complete brain coverage (including the whole cerebellum)
- Visual image check for image artifacts
- Evaluation of participant motion:

An automated estimation of translations and rotations between reference time frame #1 and time frames #2-4 was performed, using the *Motion Correction* function implemented in the PMOD Fusion module to identify time frames with significant frame displacement (> 4 mm: *QIBA Amyloid PET Profile*: [QIBA, 2022](#)).

In these rare cases, it was checked whether the time frames could be censored, affording that ≥ 2 valid time frames (= a minimum scan time of 10 min) were left. Here, the user is given the alternative choice between preprocessed datasets (see below) based on the full (=4 frame) and censored datasets.

DICOM image preprocessing for data export:

In addition to the original, attenuation-corrected PET image series, **four preprocessed versions of this dataset** are provided. While many users will probably prefer the final preprocessed version, there is the option to step back to earlier processing steps, or the “raw” original reconstruction, depending on the specific research question. This data preparation strategy was adopted from the ADNI PET preprocessing pipeline (for detailed descriptions: [Jagust et al., 2015](https://adni.loni.usc.edu/methods/pet-analysis-method/pet-analysis/); <https://adni.loni.usc.edu/methods/pet-analysis-method/pet-analysis/>). Yet, the data processing pipeline in PMOD (PMOD Technologies LLC, Zurich), which is schematically presented in Figure 1, required some technical changes:

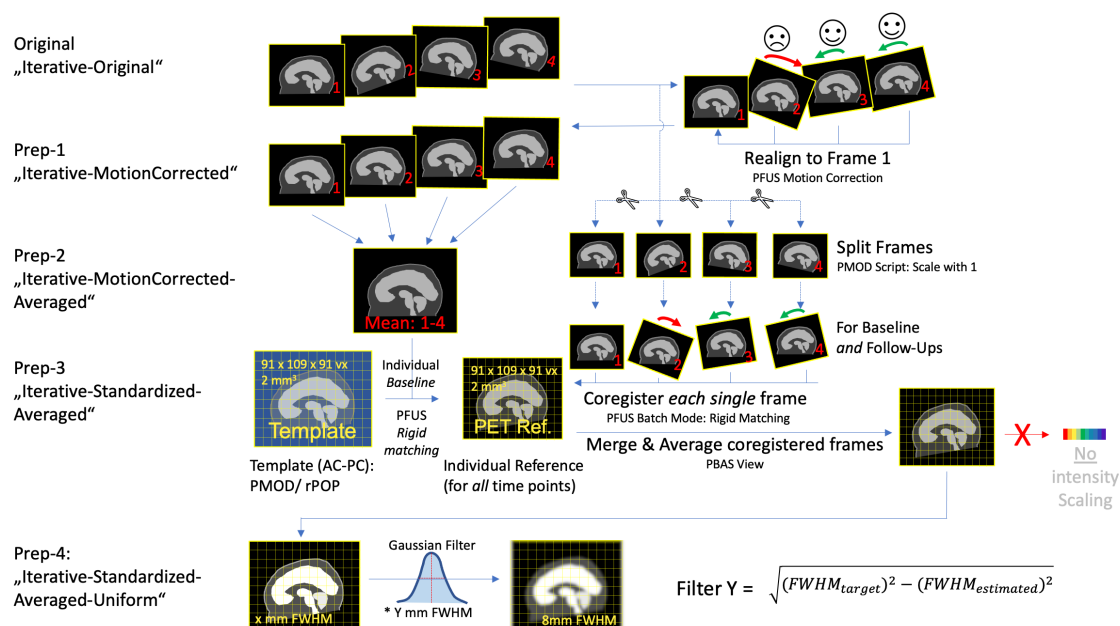


Figure 1. Schematic outline of PET data preprocessing for DELCODE

Series description	Details
Original: “Iterative-Original”	Time series with 4 frames of 5 min duration each. The attenuation-corrected, iterative reconstruction of the original PET acquisition, as uploaded by the PET sites, retaining the individual scanner-specific image geometry (i.e., reconstruction matrix, in-plane pixel size, slice thickness and number) and effective image resolution.
Prep-1: “Iterative-MotionCorrected”	Time series with 4 frames of 5 min duration each. Time frames #2 to #4 were realigned with reference time frame #1 in the PMOD Fusion Module, and resliced into a new time series. The original scanner-specific image geometry and effective image resolution is preserved.

Prep-2: <i>“Iterative-MotionCorrected-Averaged”</i>	Single time frame of 20 min duration (unless censored). Average of the motion-corrected series from Prep-1. The original scanner-specific image geometry and effective image resolution is preserved.
Prep-3: <i>“Iterative-Standardized-Averaged”</i>	Single time frame of 20 min duration (unless censored). The original PET time series were coregistered to tracer-specific templates, and resliced to derive interpolated PET images with a common, scanner-independent geometry (91 x 109 x 91 voxels, with 2 x 2 x 2 mm voxel size). Similar to ADNI, this included several processing steps: 1) For each subject, the motion-corrected, averaged image from the baseline PET scan was rigidly matched to the template (with optional manual corrections after visual inspection), and resliced into template voxel space to serve as a subject-specific reference. 2) Each single time frame of <i>both</i> baseline <i>and</i> follow-up PET scans was rigidly matched to this subject-specific reference, and resliced. As a result, each single time frame is only interpolated once, and should have a similar orientation and voxel space for a given subject, not only <i>within</i>, but also <i>across</i> baseline and follow-up scans for the respective tracer. 3) After visual checking, the coregistered frames of each PET time series were averaged. Because the original ADNI templates were not available, images were coregistered to a PMOD-custom FDG template (constructed from N=102 ADNI controls), or an FBB-specific template taken from the <i>rPOP</i> toolbox (laccarino et al., 2022: constructed from N=50 amyloid-positive and N=50 amyloid-negative subjects with FBB scans), respectively. Note that both coregistration steps (based on Normalized Mutual Information) applied <i>rigid body</i> transformations only: Similar to ADNI, no spatial normalization (e.g., to MNI template space) with affine or nonlinear warping was performed. Moreover, although the standardized images will often be roughly aligned, no explicit orientation to MNI coordinate space (e.g., setting of image origin to anterior commissure) is provided. Finally, be aware that the standardized PET series are not intensity scaled (= not “SUVR” or even “DVR” images). The respective procedures in ADNI (iterative scaling of global signal mean for FDG, scaling by cerebellar gray matter for amyloid PET) could not be reproduced satisfactorily in PMOD. <i>This means that users have to apply their own scaling</i>

	during image data analysis, if necessary. This can be achieved by explicit image scaling (e.g., dividing each voxel intensity by the mean signal of a reference region), or implicitly in statistical modelling (e.g., SPM proportional scaling).
Prep-4: <i>“Iterative-Standardized-Averaged-Uniform”</i>	Single time frame with 20 min duration (unless censored). Standardized, averaged images are smoothed with a scanner-specific Gaussian kernel, to achieve a uniform isotropic resolution of ~8³ mm FWHM (full-width-half-maximum). This is similar to the ADNI PET data, and of special interest for data analysis pipelines that include partial volume effect (PVE) correction, which usually request the entry of a filter parameter (which would be “8 mm” in the present case). Independent from voxel geometry, PET reconstructions differ regarding their effective spatial resolution (based on scanner-specific factors like detector technology, reconstruction method and post-reconstruction filter). We adopted a method from Joshi et al. (2009) to estimate spatial resolution empirically, based on 3D-Hoffman brain phantom scans with the acquisition protocols that were collected at the study sites. Basically, the unknown resolution was inferred by applying Gaussian smoothing to a digital version of the Hoffman phantom to identify the smoothing parameter settings that produced the least mismatch between the <i>measured</i> phantom and the smoothed <i>digital</i> phantom. This was implemented in a Matlab script written by Ralph Buchert (Universitätsklinik Eppendorf, Hamburg), with slight modifications. It followed the necessary processing steps described in the <i>QIBA Amyloid PET Profile</i> (QIBA, 2022: Appendix H). Because the original digital phantom was not available, a more recent, publicly available version with a higher voxel resolution was used (Harrison et al., 2020). Smoothing kernels were varied independently for the x-y and z dimensions, in 0.5 mm steps, to find the solution where subtraction of the two coregistered images resulted in minimum sum of error squares. Based on these estimates, the size of the Gaussian filter (in x-y <i>versus</i> z direction) that was necessary for a target smoothness of 8 mm FWHM was approximated, with: $\text{Filter} = \sqrt{(FWHM_{\text{target}})^2 - (FWHM_{\text{estimated}})^2}$ Again, filter sizes were rounded to the closest 0.5 mm value.

	The chosen target resolution of 8mm FWHM is compatible with ADNI, but was also the least common denominator necessary for including the scanner with the lowest resolution. <i>In principle</i>, you could drop the latter, take the unsmoothed standardized images from Prep-3, and use the formula to calculate Gaussian kernels that are necessary to smooth these PET images to an alternative resolution (e.g., in SPM). Yet, note that the majority of the available PET data had reconstructions with estimated 7 mm resolution, leaving little space for improvement without sacrificing substantial amounts of data.
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As exemplified in Figure 2, existing differences in image appearance due to the varying total fields of view, voxel sizes and intrinsic resolution are incrementally reduced by preprocessing, with the final *Prep-4* version showing harmonized image quality:

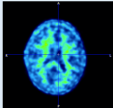
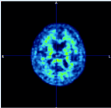
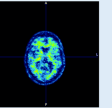
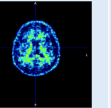
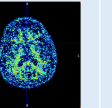
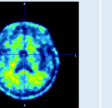
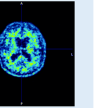
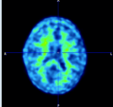
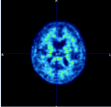
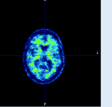
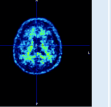
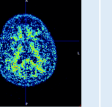
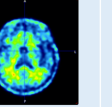
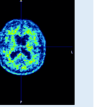
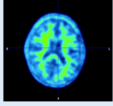
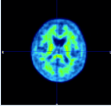
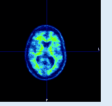
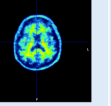
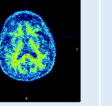
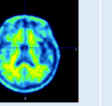
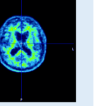
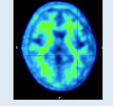
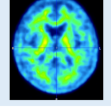
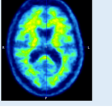
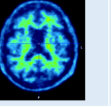
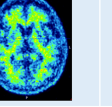
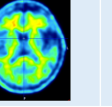
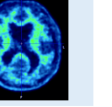
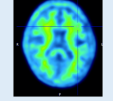
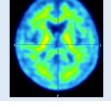
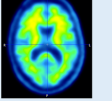
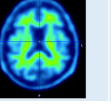
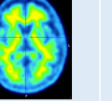
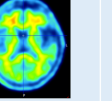
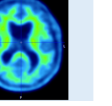
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Original							
Motion corrected							
Motion corrected + averaged							
Standardized + averaged							
Standardized + averaged + smoothed							

Figure 2. Schematic outline of the image preprocessing outcomes (Florbetaben examples)

Data formatting:

The original and preprocessed PET image data are provided in **DICOM format**. To harmonize file formatting, the original DICOM file names (which were highly variable between sources) were replaced by a **unified naming scheme** *Tracer_SeriesDescription_Img_ImageNumber* during final data export.

During export, additional **editing of original DICOM header information** was necessary. Most importantly, addresses which can potentially contain *location*- and especially *patient*-relevant information (except for the birth year [YYYY0101, *if provided*], age, sex, body length [*if provided*], and body weight) were deleted or replaced for **pseudonymization purposes**.

Since the original **DICOM series descriptions** varied widely for different vendors and scan protocols (which is quite different from the situation with the Siemens only MR protocols in DELCODE), they were replaced by **uniform Series Descriptions** (DICOM address: 0008,103E), also to simplify later DICOM conversion and data processing. Yet, the original series descriptions are written into the *Image Comments* field (DICOM address: 0020,4000) to keep the information traceable. The latter field also contains additional information about reconstruction parameters that are not reliably present in the regular DICOM addresses (e.g., number of iterations and subsets). Finally, the additional Gaussian filters that applied to the *Prep-4* series were coded here.

Note that PMOD changes or deletes some auxiliary DICOM addresses during image processing (especially: vendor-specific private tags). Although some relevant information is restored in the PMOD-generated series from the original DICOM series during final data export, the most reliable source of information is the original DICOM image series.

Additional variables:

The DICOM image data are provided with accompanying descriptive variables which go beyond DICOM header information.

FBB_DCM_COMM_P0 FDG_DCM_COMM_P0 FBB_DCM_COMM_P24 FDG_DCM_COMM_P24	<i>Quality comments</i> Mainly regarding limitations to usability, or recommendations for potential use of the regarding the PET DICOM dataset.
FBB_DATE_P0 FDG_DATE_P0 FBB_DATE_P24 FDG_DATE_P24	<i>Date of PET examination</i>
FBB_REF_P0 FDG_REF_P0 FBB_REF_P24 FDG_REF_P24	<i>Clinical reference visit</i> Especially for FBB, PET exam dates may fall in-between two consecutive DELCODE main study visits. Here, the study visit <i>closer in time</i> is suggested for reference (which can be the next one).
FBB_REFMR_P0 FDG_REFMR_P0 FBB_REFMR_P24 FDG_REFMR_P24	<i>MR reference visit (M0/ M12/ M24....)</i> Since the clinical visits not always included an MR examination, or was scheduled on a different date, the MR visit closest in time was not always identical with the clinical reference visit. Sometimes, no contemporary MR data were available, but at least from the year <i>before</i> and/ or <i>after</i> PET exam. Again, the MR visit closer in time is suggested (in some equivocal cases, both). If no recent (< ~1 year) MR was available, the field was coded "N/A". Without a recent MR image, use of MR-based pipelines is compromised, and without any MR image, not possible. Meanwhile, <i>PET-only</i> analysis pipelines which spatially normalize images with a modality-specific template are less flexible, but generally possible. While this is well-established for FDG (e.g. Della Rosa et al., 2014), complementary solutions for amyloid PET have emerged more recently (e.g. rPOP: Iaccarino et al., 2022).
AGE_FBB_P0 AGE_FDG_P0 AGE_FBB_P24 AGE_FDG_P24	<i>Age at time of PET examination</i>
FBB_DOSE_P0 FDG_DOSE_P0 FBB_DOSE_P24 FDG_DOSE_P24	<i>Decay-corrected injected tracer activity [in Megabequerel, MBq]</i> For some participants, the injected FBB tracer dose was lower than defined in the DELCODE protocol (260-300 MBq), but still consistent with manufacturer's suggestions (300±60 MBq) in most

	cases. Tracer over- and underdosings were not generally excluded, if image quality was normal, but this variable can be used for selecting participants more stringently. Note that the respective DICOM address (0018,1074) will often show different values. The information is entered manually at the scanner console, and were frequently based on a simple subtraction of syringe activity before and after injection (which is valid only with short delays between the measurement, due to the exponential decay of tracer activity). Therefore, available checklist information was used to calculate more precise estimates.
FBB_DELAY_P0 FDG_DELAY_P0 FBB_DELAY_P24 FBG_DELAY_P24	<i>Time between tracer injection and start of PET emission</i> [in min] Based on available DICOM and checklist information. For some participants, emission scan did not start regularly, i.e. 40 (FDG) or 90 min (FBB) post-injection, usually due to technical problems. While there is preliminary empirical evidence (Barthel et al., 2017) that Florbetaben SUVR values are robust across various time windows between 45 and 130 minutes, users can use this variable to select cases more stringently. Note that the DICOM information for <i>Radiopharmaceutical Start Time</i> (0018,1072) is not always reliable, since this time point is usually entered manually, and potential subject to typos. While this influences the absolute pixelwise Standardized Uptake Value (SUV) values, it should have no effect on the usual analytical approaches in brain imaging, which scale against a reference region (=effectively work with SUV ratio scores).
FDG_DIET_P0 FDG_DIET_P24	(FDG only) <i>Duration of food abstinence</i> [in hours] Duration was sometimes not reported numerically, but ordinal information was given (e.g. >6 hours = the requested minimum). Meanwhile, this was considered acceptable, since food abstinence mainly served as prerequisite for assuring the primary criterion, which was....
FDG_BGL_P0 FDG_BGL_P24	(FDG only) <i>Blood glucose level</i> [in mg/dl] From the blood glucose measurement performed immediately before FDG injection.
FDG_REST_P0 FDG_REST_P24	(FDG only) <i>Rest period after injection</i> [in min] The requested <i>minimum</i> resting period was 20 min.
FBB_PETCAM_P0 FDG_PETCAM_P0 FBB_PETCAM_P24 FDG_PETCAM_P24	<i>Vendor and model of PET scanner</i> • GE Discovery 690: PET/CT • Philips Gemini TF 16 TOF: PET/CT (@two sites) • Siemens Biograph Duo: PET/CT • Siemens Biograph mCT 128 Flow Edge: PET/CT

	Siemens Biograph mMR: PET/MRI (@two sites)
FBB_RECON_P0 FDG_RECON_P0 FBB_RECON_P24 FDG_RECON_P24	<i>Original PET reconstruction algorithm</i> Users who received DICOM image data can also find the relevant information in the DICOM header (note here the extended <i>Image Comments</i> field).
FBB_GEO_P0 FDG_GEO_P0 FBB_GEO_P24 FDG_GEO_P24	<i>Original PET image geometry</i> Users who receive DICOM image data can also find the relevant information in the DICOM header (note here the additional information in the extended DICOM <i>Image Comments</i> field).
FBB_DEVICE_P0 FDG_DEVICE_P0 FBB_DEVICE_P24 FDG_DEVICE_P24	<i>PET scanner ID</i> For optional use of scanner identity as a statistical covariate. Note that some clinical sites used the same PET scanner. Identical with <i>Device Serial Number</i> (0018,1000) in DICOM header. Because the original vendor information was already deleted by some PET sites before upload, and for pseudonymization, each single scanner received a new, arbitrary ID string ("00001"- "00007").
FBB_RESL_P0 FDG_RESL_P0 FBB_RESL_P24 FDG_RESL_P24	<i>Original PET image resolution [in mm]</i> Gaussian filter full-width-half-maximum (FWHM) for the original image reconstruction. Based on estimates that were derived from 3D-Hoffman brain phantom measurements with these reconstruction protocols.
FBB_FWHM_P0 FDG_FWHM_P0 FBB_FWHM_P24 FDG_FWHM_P24	<i>Additional PET smoothing kernel [in mm]</i> Size of the additional Gaussian kernel applied to the <i>Prep-4</i> image series to approximate an effective image resolution of 8mm FWHM
FBB_MAXTR_P0 FDG_MAXTR_P0 FBB_MAXTR_P24 FDG_MAXTR_P24	<i>Maximum translation between frames in x, y or z direction [in mm]</i> To allow a more targeted inspection of original PET time series where movement-related artefacts may play a role.

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